

BST Physics Curriculum Vol 14 Chapter 3 — Shannon Channel Capacity v0.4 (refilled per Cal #104)

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Vol 14 Chapter 3 — Shannon Channel Capacity

Chapter motivation

Shannon 1948 channel capacity theorem: discrete memoryless channel (DMC) with input X + output Y has capacity $C = \max_{\{p(X)\}} I(X; Y)$ where $I(X; Y) = H(X) - H(X | Y)$ is mutual information; achievable error-free communication rate $\leq C$; rates $> C$ have non-zero error probability asymptotically. Shannon-Hartley specialized formula for continuous bandwidth-limited channel: $C = B \cdot \log_2(1 + S/N)$ where B = bandwidth, S = signal power, N = noise power.

BST cross-link per Vol 14 Ch 1-2: substrate-tick GF(128) per Koons tick gives substrate channel capacity $\leq \log_2(128) = 7$ bits per tick = $\sim 7 \times 10^{120}$ bits/second per substrate-coordinate. Substrate noise model per T2469 SCMP (Friday): finite-bandwidth observers limit information extraction; channel capacity bounded by observer Zone-2 K-type coverage.

Section 3.0b — Reader-grade 3-level pedagogy

Level 1: Shannon 1948 channel capacity $C = \max I(X; Y)$ + Shannon-Hartley $C = B \log_2(1+S/N)$; BST substrate-tick channel capacity ≤ 7 bits per Koons tick (10^{-120} s) per substrate-coordinate; substrate noise model from T2469 SCMP finite-bandwidth observers.

Level 2 (graduate-physicist/information-theorist): Shannon 1948 (Claude Shannon, *A Mathematical Theory of Communication*, Bell System Tech J): foundational paper introducing information entropy $H(X) = -\sum p_i \log p_i$, channel capacity $C = \max_{\{p(X)\}} I(X; Y)$ with $I(X; Y) = H(X) - H(X | Y) = H(Y) - H(Y | X) = H(X) + H(Y) - H(X, Y)$ mutual information; Shannon's channel coding theorem: rate $R < C \rightarrow$ arbitrarily-small error probability achievable with coding; $R > C \rightarrow$ bounded-away-from-zero error probability. Shannon-Hartley theorem (continuous

bandwidth-limited Gaussian channel): $C = B \cdot \log_2(1 + S/N)$ where B = bandwidth (Hz), S = signal power, N = noise power. BST substrate-tick channel capacity: per Vol 14 Ch 1-2 substrate operates on GF(128) per Koons tick ($\tau_{\text{Koons}} \approx 10^{-120}$ s, T2405); substrate channel capacity per substrate-coordinate $\leq \log_2(128) = 7$ bits per tick = 7×10^{120} bits/second. For substrate-coordinate count k (per substrate-tick UV cutoff at $n_C = 5$ substrate-coordinates per T2437): total substrate channel capacity $\leq 7k \times 10^{120}$ bits/second $\leq 35 \times 10^{120}$ bits/second (substrate-tick UV ceiling). T2469 SCMP framework (Lyra Friday SP-31 Casey-named #8) provides substrate noise model: observer with finite Zone-2 K-type bandwidth B_{obs} records marginal of substrate joint distribution; channel capacity from substrate to observer bounded by B_{obs} (substrate-cartography). Sub-Tsirelson $1/2^{n_C} = 1/8 = 0.125$ (Vol 5 Ch 8 Bell-CHSH + T2469) is substrate-mediated correlation signature distinguishing BST from standard QM channel capacity. Per Vol 14 Ch 2 Reed-Solomon substrate-natural coding achieves substrate-channel capacity at sub-Singleton bound efficiency. Cross-link Vol 6 Ch 3 entropy (substrate Shannon-counting at substrate-tick) + Vol 6 Ch 12 Information Theory + Thermodynamics bridging.

Level 3 (5th-grader accessible): Shannon (1948) showed information channels have a maximum reliable communication rate called “channel capacity” C . BST identifies the substrate-tick GF(128) channel as the universe’s information channel: each Koons tick (10^{-120} seconds) carries up to 7 bits of information per substrate-coordinate (since $128 = 2^7$). Substrate channel capacity $\approx 7 \times 10^{120}$ bits/second per coordinate — fast but finite. T2469 SCMP (proven Friday) says observer bandwidth limits how much substrate information any observer can extract.

Section 3.1 — Shannon 1948 Channel Capacity Theorem

$C = \max_{\{p(X)\}} I(X; Y)$ with $I(X; Y) = H(X) - H(X | Y)$.

Shannon channel coding theorem: rate $R < C \rightarrow$ arbitrarily-small error; $R > C \rightarrow$ bounded error.

Section 3.2 — Shannon-Hartley Formula

For continuous bandwidth-limited Gaussian channel: $C = B \cdot \log_2(1 + S/N)$.

B = bandwidth (Hz); S = signal power; N = noise power.

Section 3.3 — Substrate-Tick Channel Capacity

Per Vol 14 Ch 1-2: substrate GF(128) per Koons tick ($\tau_{\text{Koons}} \approx 10^{-120}$ s, T2405).

Substrate channel capacity per coordinate $\leq \log_2(128) = 7$ bits per tick = 7×10^{120} bits/second.

Total substrate-tick channel $\leq 7k \times 10^{120}$ bits/second for k substrate-coordinates (UV cutoff at $n_C = 5$; $\leq 35 \times 10^{120}$ bits/second ceiling).

Section 3.4 — T2469 SCMP Substrate Noise Model

T2469 SCMP framework (Lyra Friday Casey-named #8): observer with finite Zone-2 K-type bandwidth B_{obs} records marginal of substrate joint distribution.

Channel capacity substrate→observer bounded by B_{obs} (substrate-cartography reading).

Section 3.5 — Sub-Tsirelson 1/8 BST Substrate Signature

Per Vol 5 Ch 8 + T2469: substrate-mediated correlations exhibit sub-Tsirelson $1/2^N c = 1/8 = 0.125$ deviation.

BST distinction from standard QM channel capacity; operational falsifier at SP-30 Bell experiment design \$300-500K.

Section 3.6 — Reed-Solomon Substrate-Natural Capacity

Per Vol 14 Ch 2: Reed-Solomon on GF(128) achieves substrate-channel capacity at Singleton bound efficiency (MDS code).

Section 3.7 — Honest scope + Connection

- Shannon 1948 channel capacity [?]
- Substrate-tick capacity $\sim 7 \times 10^{120}$ bits/s per coordinate [?]
- T2469 SCMP substrate noise model [?]
- Open scope: detailed substrate-Shannon-capacity precision derivation (multi-week)

Connection: - Vol 14 Ch 1 Substrate as Information Channel - Vol 14 Ch 2 Reed-Solomon Coding - Vol 5 Ch 8 Bell-CHSH (1/8 sub-Tsirelson) - Vol 6 Ch 12 Information Theory + Thermodynamics

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